File and Directory in Practice

Computer System III

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Two Key Abstractions

- File
  - A linear array of bytes, with each you can read/write
  - Has a low-level name (user does not know) - **inode number**
  - Usually OS does not know the exact type of the file
- Directory
  - Has a low-level name
  - Its content is quite specific: contains a list of user-readable name to low-level name. Like (“foo”, inode number 10)
  - Each entry either points to file or other directory

Two Key Abstractions

- File
  - External name (visible to user) must be symbolic
    - In hierarchical file system, unique external names are given as pathnames (path from root to the file)
  - Internal names (low-level names): i-node in UNIX
    - Stores information about file system object: file, directory, socket, …
    - Index into array of file descriptors/headers for a volume
- Directory: translation from external to internal name

File System Interface

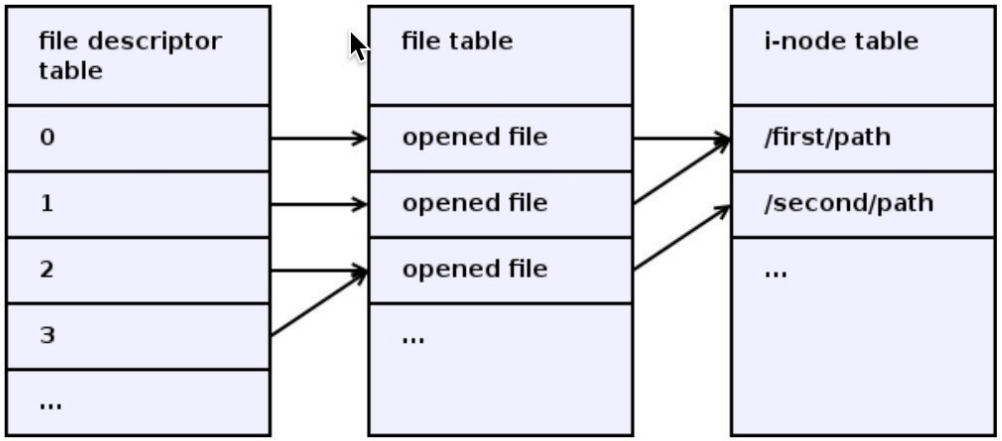
- Create file

```
int fd = open("foo", O_CREAT | O_WRONLY | O_TRUNC);
```

  - O_CREAT: create the file if not exists (not ***O_CREATE***)
  - O_WRONLY: can only write to the file
  - O_TRUNC: if the file exists, truncate it to zero bytes

The return value of **open**() is a file descriptor, a small, nonnegative integer that is used in subsequent system calls (**read**(2), **write**(2), **lseek**(2), **fcntl**(2), etc.) to refer to the open file. The file descriptor returned by a successful call will be the lowest-numbered file descriptor not currently open for the process.

File Descriptor



A *file descriptor* is an index in the per-process file descriptor table (in the left of the picture). Each file descriptor table entry contains a reference to a *file object*, stored in the file table (in the middle of the picture). Each file object contains a reference to an i-node, stored in the i-node table (in the right of the picture).

A file descriptor is just a number that is used to refer a file object from the user space. A file object represents an opened file. It contains things likes current read/write offset, non-blocking flag and another non-persistent state. An i-node represents a filesystem object. It contains things like file meta-information (e.g. owner and permissions) and references to data blocks.

File descriptors created by several `open()` calls for the same file path point to different file objects, but these file objects point to the same i-node. Duplicated file descriptors created by `dup2()` or `fork()` point to the same file object.

A BSD lock and an Open file description lock is associated with a file object, while a POSIX record lock is associated with an `[i-node, pid]` pair. We'll discuss it below.

One example

- Stdin, stdout, stderr

```
wenbo@parallels:~/os-course$ strace ./a.static
execve("./a.static", ["./a.static"], 0x7ffc89bb12f0 /* 68 vars */) = 0
brk(NULL)                                       = 0xc17000
brk(0xc181c0)                                   = 0xc181c0
arch_prctl(ARCH_SET_FS, 0xc17880)               = 0
uname({sysname="Linux", nodename="parallels", ...}) = 0
readlink("/proc/self/exe", "/home/wenbo/os-course/a.static", 4096) = 30
brk(0xc391c0)                                   = 0xc391c0
brk(0xc3a000)                                   = 0xc3a000
access("/etc/ld.so.nohwcap", F_OK)              = -1 ENOENT (No such file or directory)
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(136, 2), ...}) = 0
write(1, "hello world!\n", 13hello world!
)           = 13
exit_group(0)                                   = ?
+++ exited with 0 +++
```

The cat example

- strace cat main.c
- What is the file descriptor of main.c?

```
openat(AT_FDCWD, "main.c", O_RDONLY)    = 3
newfstatat(3, "", {st_mode=S_IFREG|0674, st_size=89, ...}, AT_EMPTY_PATH)
fadvise64(3, 0, 0, POSIX_FADV_SEQUENTIAL) = 0
mmap(NULL, 139264, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0)
read(3, "#include <stdio.h>\n\nint main () "..., 131072) = 89
write(1, "#include <stdio.h>\n\nint main () "..., 89) = 89
read(3, "", 131072)                     = 0
munmap(0xffff969f7000, 139264)          = 0
close(3)                                = 0
close(1)                                = 0
close(2)                                = 0
exit_group(0)                           = ?
+++ exited with 0 +++
```

write and fsync

- write():
  - Write this data to persistent storage, at some point in the future
  - The file system, for performance reasons, will buffer such writes in memory for some time (say 5 seconds, or 30); at that later point in time, the write(s) will actually be issued to the storage device
- fsync(): forces all dirty (not yet written) data to disk

```
int fd = open ("foo", _CREAT | O_WRONLY | O_TRUNC);
assert (fd > -1);

int rc = write (fd, buffer, size);
assert (rc == size) ;

rc = fsync (fd) ;
assert (rc == 0) ;
```

Getting Information about Files

- Information is kept in **struct stat**

```
24  struct stat {
25          unsigned long   st_dev;         /* Device.  */
26          unsigned long   st_ino;         /* File serial number.  */
27          unsigned int    st_mode;        /* File mode.  */
28          unsigned int    st_nlink;       /* Link count.  */
29          unsigned int    st_uid;         /* User ID of the file's owner.  */
30          unsigned int    st_gid;         /* Group ID of the file's group. */
31          unsigned long   st_rdev;        /* Device number, if device.  */
32          unsigned long   __pad1;
33          long            st_size;        /* Size of file, in bytes.  */
34          int             st_blksize;     /* Optimal block size for I/O.  */
35          int             __pad2;
36          long            st_blocks;      /* Number 512-byte blocks allocated. */
37          long            st_atime;       /* Time of last access.  */
38          unsigned long   st_atime_nsec;
39          long            st_mtime;       /* Time of last modification.  */
40          unsigned long   st_mtime_nsec;
41          long            st_ctime;       /* Time of last status change.  */
42          unsigned long   st_ctime_nsec;
43          unsigned int    __unused4;
44          unsigned int    __unused5;
45  };
```

```
os@os:~/temp$ stat foo
  File: 'foo'
  Size: 6               Blocks: 8          IO Block: 4096   regular file
Device: 801h/2049d      Inode: 1328649     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/      os)   Gid: ( 1000/      os)
Access: 2018-12-19 00:24:02.448286431 +0800
Modify: 2018-12-19 00:24:01.316296543 +0800
Change: 2018-12-19 00:24:01.316296543 +0800
 Birth: -
```

Removing Files

- Why we just “remove” or “delete” the file, but using “unlinkat”?

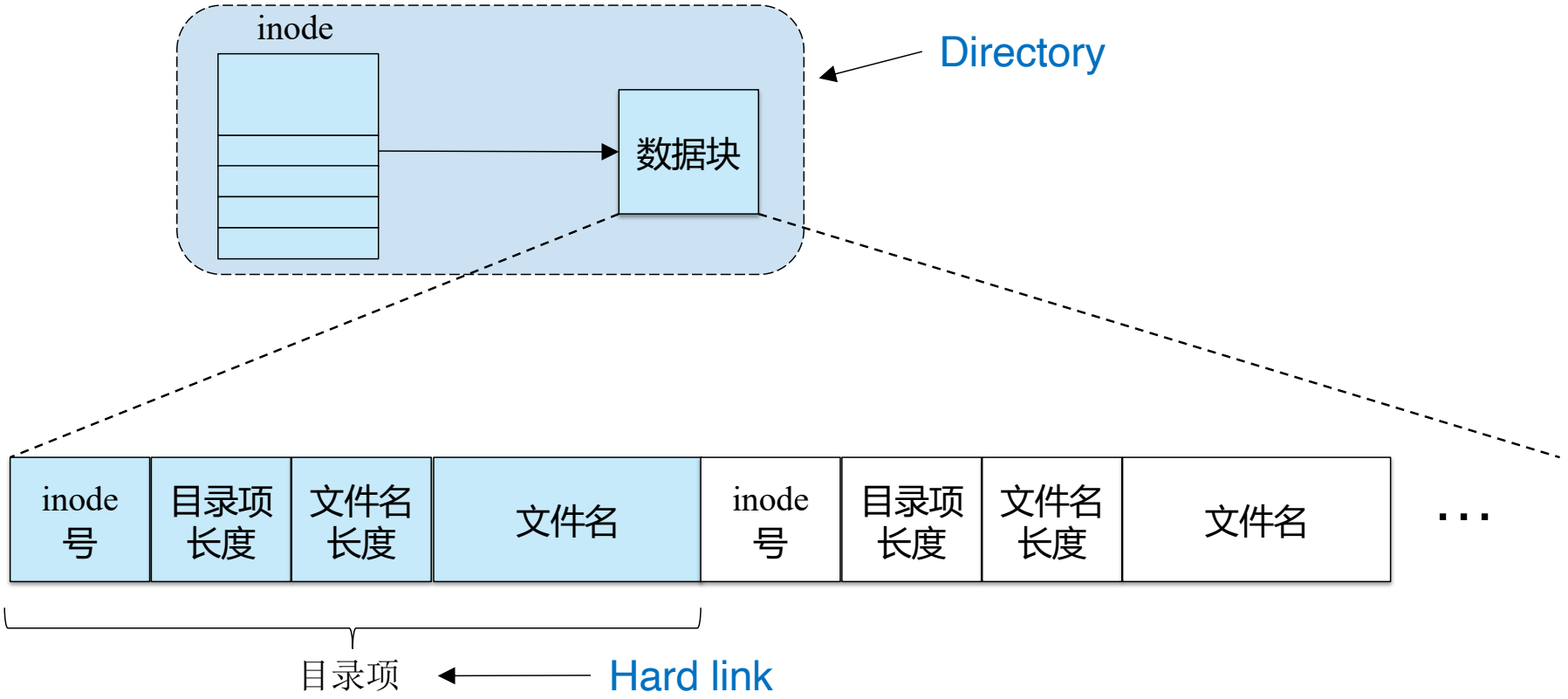
```
parallels@ubuntu-linux-22-04-02-desktop:~/os-course$ strace rm tmp
execve("/usr/bin/rm", ["rm", "tmp"], 0xffffc5c6f3a8 /* 54 vars */) = 0
brk(NULL)                               = 0xaaaae84c7000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=63887, ...}, AT_EMPTY_PATH)
mmap(NULL, 63887, PROT_READ, MAP_PRIVATE, 3, 0) = 0xffff9bb95000
close(3)                                = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
newfstatat(AT_FDCWD, "tmp", {st_mode=S_IFREG|0664, st_size=89, ...}, AT_SYMLI
faccessat2(AT_FDCWD, "tmp", W_OK, AT_EACCESS) = 0
unlinkat(AT_FDCWD, "tmp", 0)            = 0
lseek(0, 0, SEEK_CUR)                   = -1 ESPIPE (Illegal seek)
close(0)                                = 0
close(1)                                = 0
close(2)                                = 0
exit_group(0)                           = ?
+++ exited with 0 +++
```

Hard link vs soft link

- Link
  - A file may be known by more than one name in one or more directories. Such multiple names are known as links.
  - Two kinds of links are also known as hard links and soft links.
- Hard link
  - A hard link is a directory entry that associates with a file
  - The file name "." in a directory is a hard link to the directory itself
  - The file name ".." is a hard link to the parent directory
- Soft link (a.k.a., Symbolic link or symlink)
  - A symbolic link is a file containing the path name of another file
  - Soft links, unlike hard links, may point to directories and may cross file-system boundaries

Hard link

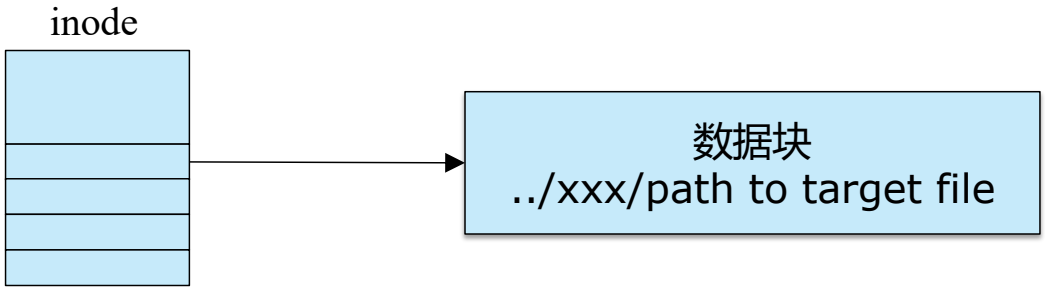
- Directory is a file
- A hard link is a directory entry




```
struct ext2_dir_entry {
        __le32  inode;                  /* Inode number */
        __le16  rec_len;                /* Directory entry length */
        __le16  name_len;               /* Name length */
        char    name[];                 /* File name, up to EXT2_NAME_LEN */
};
```

Soft link

- A soft link is a file containing the path name of another file



Hard link

- link() system call: system call takes two arguments, an old pathname and a new one
  - Another way to refer to the same file
  - Same inode number

```
parallels@ubuntu:~/os-course$ echo hello > file1
parallels@ubuntu:~/os-course$ cat file1
hello
parallels@ubuntu:~/os-course$ ln file1 file2
parallels@ubuntu:~/os-course$ ls -l file*
-rw-rw-r-- 2 parallels parallels 6 Dec 17 21:02 file1
-rw-rw-r-- 2 parallels parallels 6 Dec 17 21:02 file2
parallels@ubuntu:~/os-course$ cat file2
hello
parallels@ubuntu:~/os-course$ ls -i file1 file2
3670573 file1  3670573 file2
```

Hard link

- link() system call: system call takes two arguments, an old pathname and a new one
  - Even an empty directory has two entries
    - Which two?

```
parallels@ubuntu:~/os-course$ mkdir dir1
parallels@ubuntu:~/os-course$ ls -al dir1
total 8
drwxrwxr-x  2 parallels parallels 4096 Dec 17 21:05 .
drwxrwxr-x 20 parallels parallels 4096 Dec 17 21:05 ..
parallels@ubuntu:~/os-course$ stat dir1
  File: dir1
  Size: 4096            Blocks: 8          IO Block: 4096   directory
Device: 802h/2050d      Inode: 3803181     Links: 2
Access: (0775/drwxrwxr-x)  Uid: ( 1000/parallels)   Gid: ( 1000/parallels)
Access: 2023-12-17 21:05:59.720000277 +0800
Modify: 2023-12-17 21:05:53.220000274 +0800
Change: 2023-12-17 21:05:53.220000274 +0800
 Birth: 2023-12-17 21:05:53.220000274 +0800
```

Why removing a file calls unlink

- Create a file
  - making a structure (the inode) that will track virtually all relevant information about the file
  - **linking** a human-readable name to that file (or inode), and putting that link into a directory
- To remove a file, we **unlink** it

```
unlink() deletes a name from the filesystem.  If that name was the last
link to a file and no processes have the file open, the file is deleted
and the space it was using is made available for reuse.
```

Reference count

- rm: unlink the file, and check the reference count of the **inode**

```
parallels@ubuntu:~/os-course$ ll file1
-rw-rw-r-- 1 parallels parallels 6 Dec 17 21:02 file1
parallels@ubuntu:~/os-course$ ln file1 file2
parallels@ubuntu:~/os-course$ ll file1
-rw-rw-r-- 2 parallels parallels 6 Dec 17 21:02 file1
parallels@ubuntu:~/os-course$ ln file1 file3
parallels@ubuntu:~/os-course$ ll file1
-rw-rw-r-- 3 parallels parallels 6 Dec 17 21:02 file1
parallels@ubuntu:~/os-course$ ll file3
-rw-rw-r-- 3 parallels parallels 6 Dec 17 21:02 file3
parallels@ubuntu:~/os-course$ rm file3
parallels@ubuntu:~/os-course$ ll file1
-rw-rw-r-- 2 parallels parallels 6 Dec 17 21:02 file1
parallels@ubuntu:~/os-course$ rm file2
parallels@ubuntu:~/os-course$ ll file1
-rw-rw-r-- 1 parallels parallels 6 Dec 17 21:02 file1
```

Soft links

- Hard link number does not increase
- Different inode
- rm the target file, the link will be invalided

```
parallels@ubuntu:~/os-course$ cat file1
hello
parallels@ubuntu:~/os-course$ ln -s file1 file2
parallels@ubuntu:~/os-course$ ll file*
-rw-rw-r-- 1 parallels parallels 6 Dec 17 21:02 file1
lrwxrwxrwx 1 parallels parallels 5 Dec 17 21:14 file2 -> file1
parallels@ubuntu:~/os-course$ ls -i file*
3670573 file1  3670575 file2
parallels@ubuntu:~/os-course$ rm file1
parallels@ubuntu:~/os-course$ ll file*
lrwxrwxrwx 1 parallels parallels 5 Dec 17 21:14 file2 -> file1
```

Two different aspects of FS

- VFS data structure
  - inode
- In-memory data structure
  - ext2_inode_info
  - Contain both ext2_inode and inode
- On-disk data structure
  - ext2_inode

```
struct inode {
        umode_t                 i_mode;
        unsigned short          i_opflags;
        kuid_t                  i_uid;
        kgid_t                  i_gid;
        unsigned int            i_flags;

#ifdef CONFIG_FS_POSIX_ACL
        struct posix_acl        *i_acl;
        struct posix_acl        *i_default_acl;
#endif

        const struct inode_operations   *i_op;
        struct super_block      *i_sb;
        struct address_space    *i_mapping;

#ifdef CONFIG_SECURITY
        void                    *i_security;
#endif
```

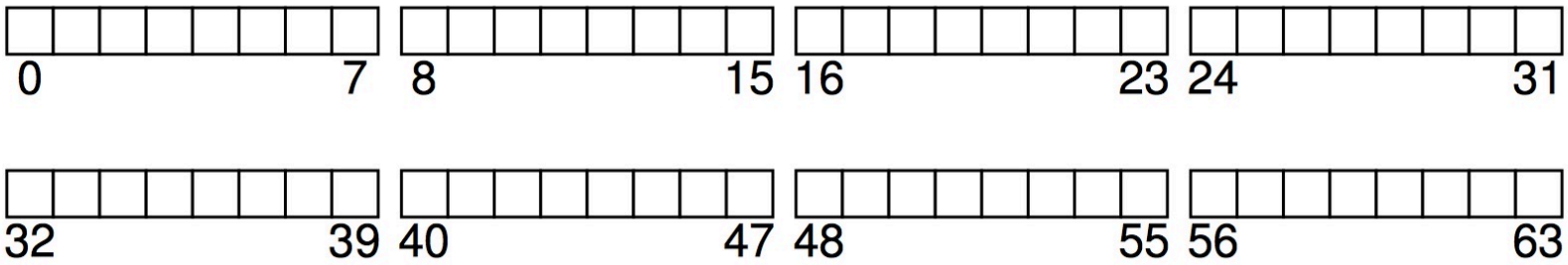
```
struct ext2_inode_info {
        __le32  i_data[15];
        __u32   i_flags;
        __u32   i_faddr;
        __u8    i_frag_no;
        __u8    i_frag_size;
        __u16   i_state;
        __u32   i_file_acl;
        __u32   i_dir_acl;
        __u32   i_dtime;

        struct mutex truncate_mutex;
        struct inode    vfs_inode;
        struct list_head i_orphan;      /* unlinked but op
#ifdef CONFIG_QUOTA
        struct dquot *i_dquot[MAXQUOTAS];
#endif
};
```

Two different aspects of FS

- (on-disk) Structure about FS
  - vs in-memory structure about FS
- Access methods
  - how does the FS maps the calls (open/read/write) onto its structures
    - inode->i_fops

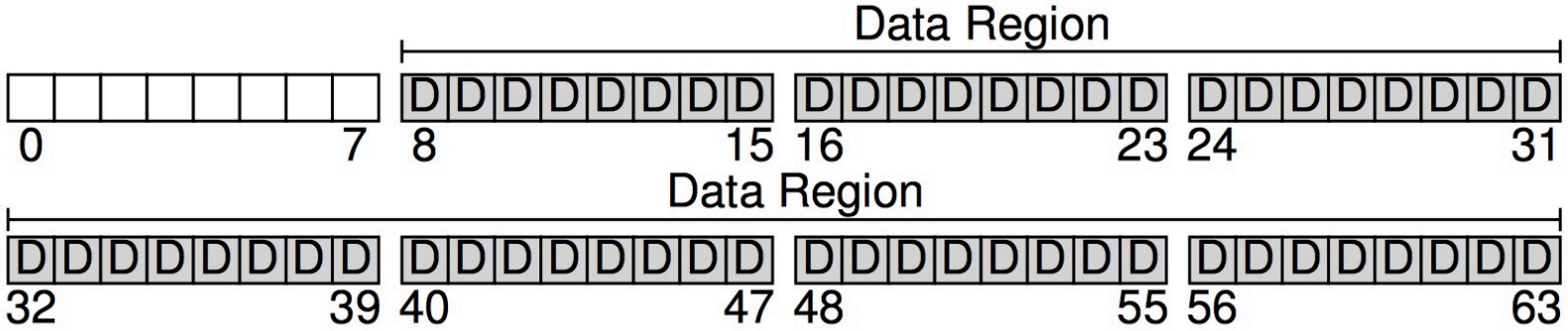
An Example of FS Organization

- Suppose we have a serial of blocks
  - Block size: 4KB
  - 64 blocks



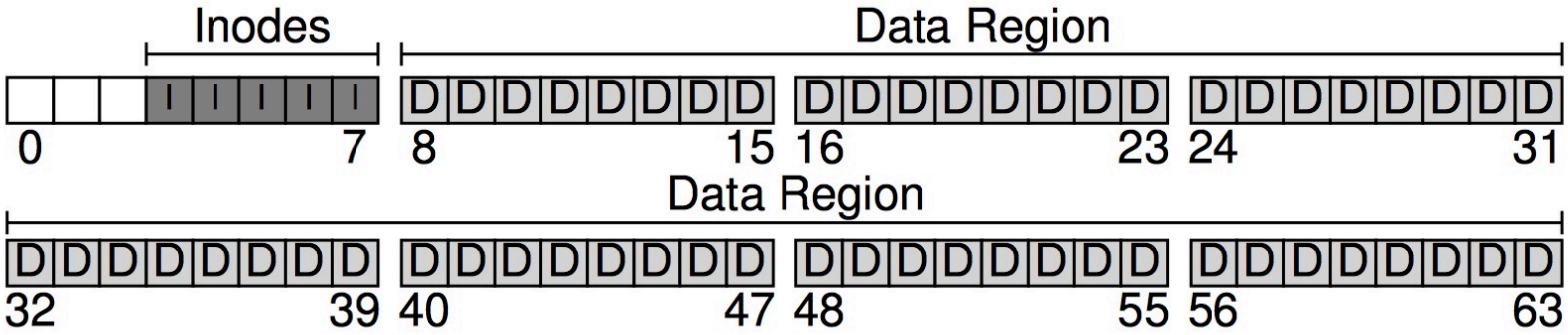
Data Region

- We reserve some blocks for data
  - 56 of 64 blocks



Inodes

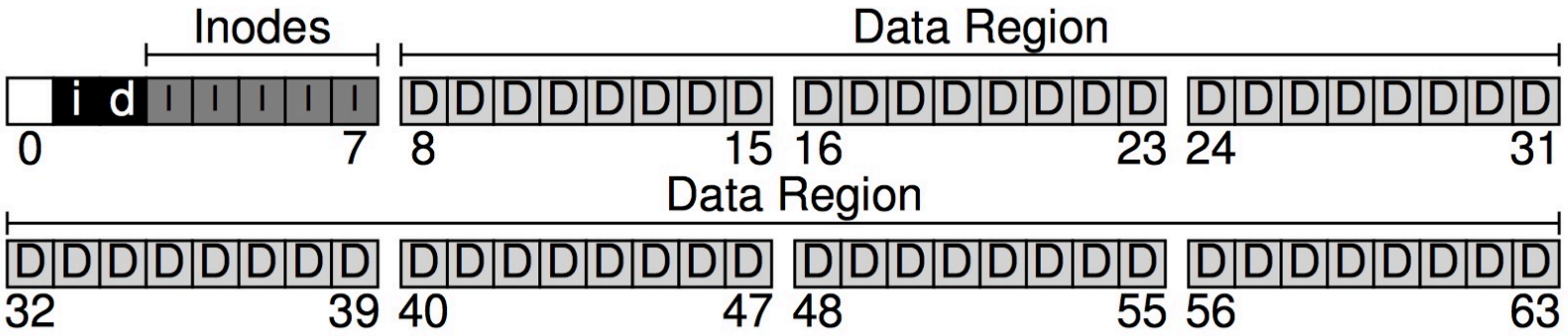
- inode table: contains inodes
- 5 of 64 blocks are reserved for inodes
  - Suppose inodes are 256 bytes, 4 KB block can hold 16 inodes, then 5 blocks -> 80 inodes -> 80 files (directories)



D: data block
I: inode table

Bitmap

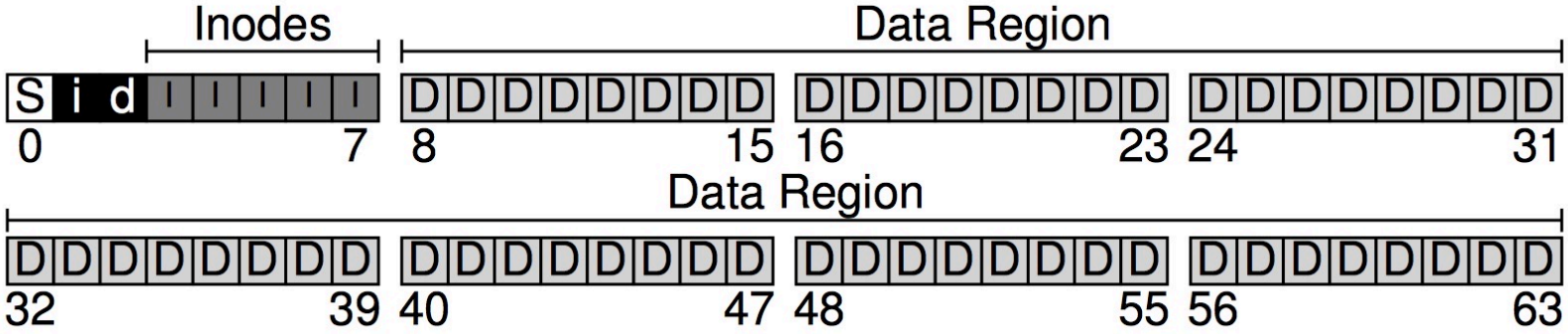
- Suppose we use bitmap to manage the free space
  - One bitmap for free inodes
  - One bitmap for free data region



D: Data block
I: inode
i: inode bitmap
d: data region bitmap

Superblock

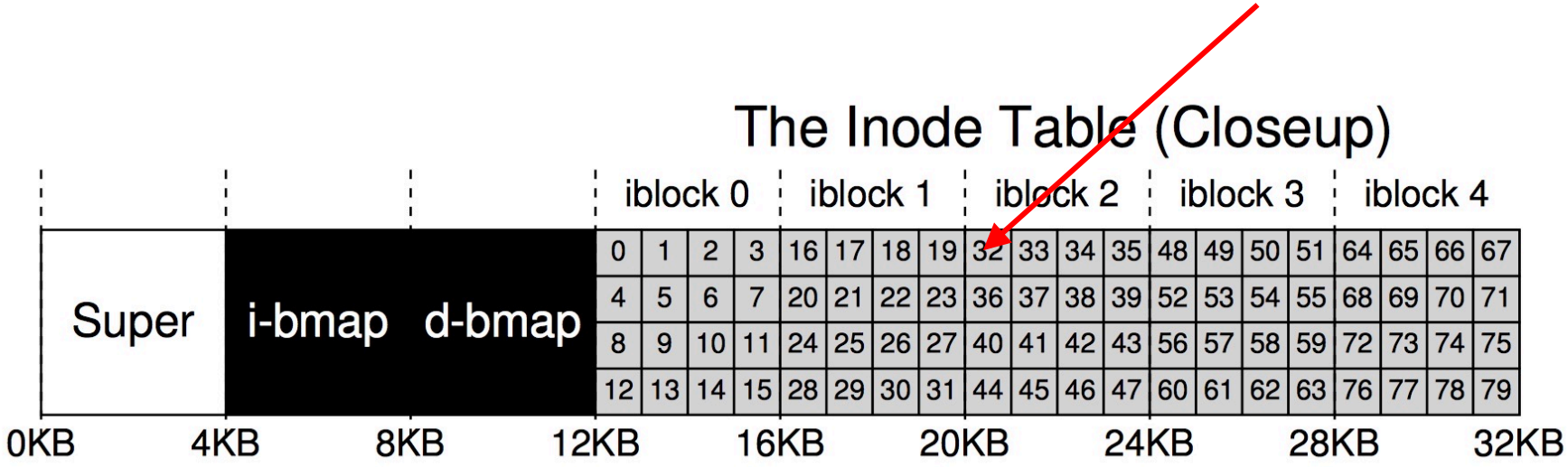
- Superblock
  - Contains information about this file system: how many inodes/data blocks, where the inode table begins, where the data region begins, and **a magic number**



D: Data block
I: inode
i: inode bitmap
d: data region bitmap
S: superblock

inode address calculation

- Suppose inodes are 256 bytes, 4 KB block can hold 16 inodes, then 5 blocks -> 80 inodes -> 80 files (directories)
- Each inode is identified by a number(inumber)
- To read inode number No. 32
  - 32 * sizeof(inode) = 8KB
  - Address: 4k(super block) + 8k(BITMAP) + 8k = 20KB



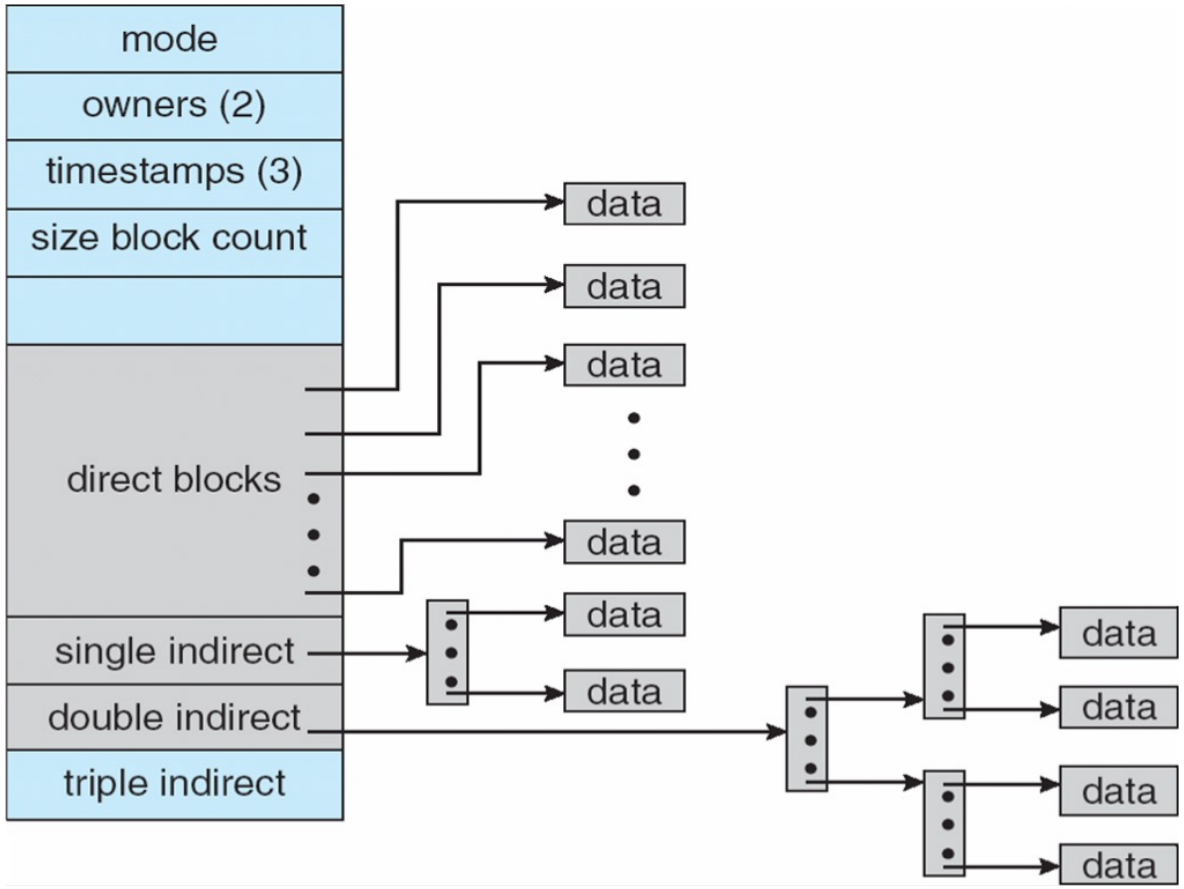
Linux ext2_inode

Size	Name	What is this inode field for?
2	mode	can this file be read/written/executed?
2	uid	who owns this file?
4	size	how many bytes are in this file?
4	time	what time was this file last accessed?
4	ctime	what time was this file created?
4	mtime	what time was this file last modified?
4	dtime	what time was this inode deleted?
2	gid	which group does this file belong to?
2	links_count	how many hard links are there to this file?
4	blocks	how many blocks have been allocated to this file?
4	flags	how should ext2 use this inode?
4	osd1	an OS-dependent field
60	block	a set of disk pointers (15 total)
4	generation	file version (used by NFS)
4	file_acl	a new permissions model beyond mode bits
4	dir_acl	called access control lists

Figure 40.1: **Simplified Ext2 Inode**

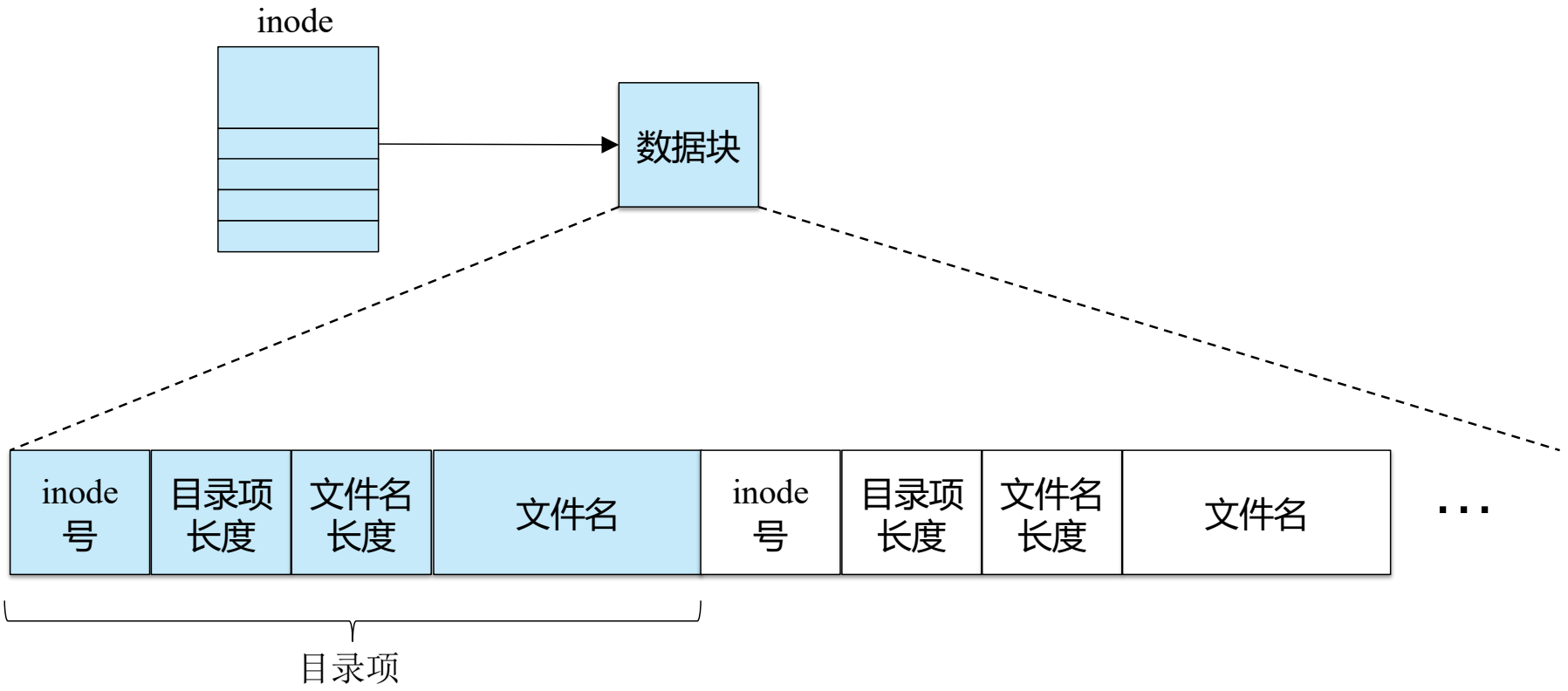
Multi-Level index in inode

- To support bigger file, we need multi-level index for the nodes



Directory Implementation

- Directory is a special file
  - Store the mapping from file name to its inode



Read /foo/bar

- Suppose foo is a directory, bar is a file in it
- / + foo/ + bar

	data bitmap	inode bitmap	root inode	foo inode	bar inode	root data	foo data	bar data[0]	bar data[1]	bar data[2]
open(bar)			**read**			**read** ← Read dir entry of foo				
				read			**read** ← Read dir entry of bar			
					read					
read()					**read**					
					write	← Update access time				
read()					**read**					
					write					
read()					**read**					
					write					**read**

Write to Disk: /foo/bar

	data bitmap	inode bitmap	root inode	foo inode	bar inode	root data	foo data	bar data[0]	bar data[1]	bar data[2]
create (/foo/bar)		read write	read	read	write	read	read	write		
write()	read write				read write				write	
write()	read write				read write					write
write()	read write				read write					

Find bar not exist

Write dir entry of bar to foo

Update metadata, such as count, access time

Caching and Buffering

- Without caching, each file open would require two reads for each level of the directory
  - One for the inode, and one for data
- Early system allocate a **fixed-size** cache to hold popular blocks
- Modern systems use a **unified page cache** for both virtual memory pages and file system pages
- Write buffering: does not write to disk immediately, instead sync to disk for like 5 - 30 seconds
- Database: direct IO with raw data

Takeaway

- File interfaces
  - External name
  - Internal name (low-level name): inode
- Directory
  - Translate from external name to internal name
- Hard link vs soft link
- On-disk layout of FS